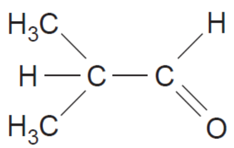
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		it is acting as a base / it removes protons			1	1		
		(ii)		nucleophilic substitution	1			1		
		(iii)	I	M_r of —CHO fragment is 29 ⇨ M_r of R is 72 – 29 = 43 (1) R is C_3H_7 but branched ⇨ compound **S** is  (1)		1	1	2		
			II	orange / red	1			1		1
			III	mixture melts at a lower temperature and over a greater range			1	1		1
	(b)	(i)		use a **separating / dropping funnel** and run off the lower aqueous layer	1			1		1
		(ii)		NaOH / I_2 or KI / NaOCl (1) yellow solid (1)	2			2		2

© WJEC CBAC Ltd.

<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td></td><td></td><td>(iii)</td><td></td><td>award (1) for each correct column
<table>
<tr><th>Protons</th><th>Splitting pattern</th><th>Relative peak area</th></tr>
<tr><td>CH_3CO</td><td>singlet</td><td>3 (or 1)</td></tr>
<tr><td>$(CH_3)_3C$</td><td>singlet</td><td>9 (or 3)</td></tr>
</table></td><td></td><td>2</td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(iv)</td><td>I</td><td>the alcohol has hydrogen bonding (the others do not) (1)

therefore more energy is needed to separate the molecules (hence higher boiling temperature) (1)</td><td></td><td>2</td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td>II</td><td>similar – they each show four separate signals (four environments) (1)

different – the (four) signals will be in different positions (1)

accept any other sensible answers</td><td></td><td>2</td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td>Question 8 total</td><td>5</td><td>7</td><td>3</td><td>15</td><td>0</td><td>5</td></tr>
</table>

© WJEC CBAC Ltd.